

Chapter 2

Equations of First Order and First Degree

2.1 Introduction: Existence Theorem

An ordinary differential equation of first order and first degree can be written as

$$\frac{dy}{dx} = f(x, y) \quad (2.1.1)$$

or in the more symmetric form

$$M dx + N dy = 0 \quad (2.1.2)$$

where $M = M(x, y)$ and $N = N(x, y)$ are functions of x and y . The following theorem holds for the general first order equation (2.1.1).

Existence and Uniqueness Theorem. *If the function $f(x, y)$ be continuous in a domain containing the point (x_0, y_0) at its interior, then there exists a function $y = \phi(x)$ satisfying equation (2.1.1) and taking the value y_0 for $x = x_0$. If, in addition, the partial derivative $\frac{\partial f}{\partial y}$ is continuous, this solution of the equation is unique.*

[C.H. 1993, '95, 2001, '03]

The condition that $y = \phi(x)$ has the value y_0 for $x = x_0$ is known as the *initial condition* of the equation.

Geometrically, the theorem asserts the existence of a unique integral curve of the equation passing through (x_0, y_0) .

Unfortunately it is not possible to obtain in all cases the general solution in terms of known functions; in many cases one is obliged to have a recourse to the method of numerical approximation. We shall, however, discuss certain simple classes in which by the application of elementary processes the general solutions can be found in terms of known functions. These classes are classified as:

1. Exact equations.
2. Equations solvable by separation of variables.
3. Homogeneous equations.
4. Linear equations of first order.

2.2 Exact Equations

A differential equation (2.1.1) of first order and of first degree can be expressed in the form

$$M(x, y) dx + N(x, y) dy = 0, \text{ or, simply, } M dx + N dy = 0$$

where M and N are functions of x and y .

If the differential $M dx + N dy$ is immediately, that is without multiplication by any factor, expressible in the form du , where u is a function of x and y , it is said to be exact and the equation $M dx + N dy = 0$ is then called an exact differential equation.

If the equation $M dx + N dy = 0$ is exact, i.e., if a function $u(x, y)$ exists such that $du = M dx + N dy$, then the general solution of the equation is, clearly, $u = c^1$, where c is an arbitrary constant.

► **Example 2.2.1** The equation $y dx + x dy = 0$ is an exact differential equation, since $y dx + x dy = d(xy)$. Its general solution is $xy = c$.

► **Example 2.2.2** $2x^2y dy + 2y^2x dx = 0$ is an exact differential equation, since $2x^2y dy + 2y^2x dx = d(x^2y^2)$. Its primitive is $x^2y^2 = c$.

Theorem 2.2.1 The necessary and sufficient condition that

$$M dx + N dy = 0 \quad (2.2.1)$$

be exact is

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}. \quad (2.2.2)$$

[C.H. 1998]

¹Throughout this text the letter c or C denotes an arbitrary constant. Any other use of these letters will be evident from the context.

Then

$$\frac{\partial F}{\partial x} = M$$

$$\text{and} \quad \frac{\partial^2 F}{\partial x \partial y} = \frac{\partial^2 F}{\partial y \partial x} = \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

So

$$\frac{\partial}{\partial x} \left(N - \frac{\partial F}{\partial y} \right) = 0.$$

$\therefore N - \frac{\partial F}{\partial y}$ is constant so far as x is concerned, that is, a function of $y = \phi(y)$, say.

Hence

$$N = \phi(y) + \frac{\partial F}{\partial y}.$$

We next take function $u(x, y)$ so that

$$u = F + \int \phi(y) dy.$$

Then

$$\frac{\partial u}{\partial y} = N.$$

Also

$$\begin{aligned} M &= \frac{\partial F}{\partial x} \quad (\text{by our construction of } F) \\ &= \frac{\partial u}{\partial x} \quad \left(\because u = F + \int \phi(y) dy \right). \end{aligned}$$

Thus,

$$M dx + N dy = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy = du,$$

a perfect differential.

So the equation (2.2.1) is exact and the condition (2.2.2) is, therefore, sufficient.

► **Example 2.2.3** Is the equation

$$(x^2 - y) dx + (y^2 - x) dy = 0$$

an exact equation?

Solution: Here $M = x^2 - y$, $N = y^2 - x$. Then

$$\frac{\partial M}{\partial y} = -1 = \frac{\partial N}{\partial x},$$

i.e., the condition of exactness is satisfied. Hence the equation is exact.

When the condition of integrability is satisfied, the *general solution* (or the *primitive*) can be obtained by the following rules:

- ✓ 1. Integrate $M dx$ as if y is constant.
- ✓ 2. Integrate $N dy$ as if x is constant but, then *delete those terms already obtained in 1.*
- ✓ 3. The sum of these integrals will give a function $u(x, y)$ such that $du = M dx + N dy$.
[The Justification of these rules can be found in the proof of the sufficiency part of Theorem 2.2.1]
- ✓ 4. Once the function u is obtained, then $u = c$ is the primitive.

► **Example 2.2.4** Solve: $(x^2 - y) dx + (y^2 - x) dy = 0$.

Solution: The equation is exact (see Example 2.2.3). We now apply the rules to find u such that $du = (x^2 - y) dx + (y^2 - x) dy$.

1. $\int (x^2 - y) dx = \frac{1}{3}x^3 - xy$ (as if y is constant).
2. $\int (y^2 - x) dy = \frac{1}{3}y^3 - xy$ (as if x is constant).

$$3. u = \left(\frac{1}{3}x^3 - xy \right) + \frac{1}{3}y^3$$

[deleting the term $-xy$ from 2 which has already been obtained in 1].

4. The required primitive is

$$\frac{1}{3}(x^3 + y^3) - xy = c$$

$$\text{or, } x^3 + y^3 - 3xy = k \text{ (where } 3c = k).$$

Important Differentials

The following differentials are easily seen to be exact:

- ✓ (a) $dx \pm dy = d(x \pm y)$
- ✓ (b) $x dx + y dy = d\left(\frac{1}{2}x^2 + \frac{1}{2}y^2\right)$

$$(c) \quad x^2 dx + y^2 dy = d \left(\frac{1}{3} x^3 + \frac{1}{3} y^3 \right).$$

(d) $X dx + Y dy = d \left\{ \int X dx + \int Y dy \right\}$, where X is a function of x alone and Y is a function of y alone.

Sometimes instead of using the condition of integrability an equation may be seen to be exact after a regrouping of its terms.

► **Example 2.2.5** Solve: $(\cos y + y \cos x) dx + (\sin x - x \sin y) dy = 0$.

Solution: Method 1. Using the condition of integrability, we find

$$\frac{\partial M}{\partial y} = -\sin y + \cos x = \frac{\partial N}{\partial x}.$$

Hence the equation is exact.

$$\int M dx = \int (\cos y + y \cos x) dx = x \cos y + y \sin x$$

(assuming y to be constant)

$$\int N dy = \int (\sin x - x \sin y) dy = y \sin x + x \cos y$$

(assuming x to be constant).

Both the terms in $\int N dy$ are already obtained in $\int M dx$.

$$\therefore u = x \cos y + y \sin x$$

and hence the primitive is $x \cos y + y \sin x = c$.

Method 2. $(\cos y + y \cos x) dx + (\sin x - x \sin y) dy = 0$.

Regrouping the terms we write

$$(\cos y dx - x \sin y dy) + (y \cos x dx + \sin x dy) = 0$$

or, $d(x \cos y) + d(y \sin x) = 0$.

Integrating, $x \cos y + y \sin x = c$, is the required primitive.

► **Example 2.2.6** Solve: $(y^2 e^{xy^2} + 4x^3) dx + (2xy e^{xy^2} - 3y^2) dy = 0$.

[C.H. 1998]

Solution: As in the previous example, the equation is exact, since

$$\frac{\partial M}{\partial y} = 2ye^{xy^2} + 2xy^3e^{xy^2} = \frac{\partial N}{\partial x}.$$

$$\therefore \int M dx = \int (y^2 e^{xy^2} + 4x^3) dx = e^{xy^2} + x^4, \quad y = \text{constant}.$$

$$\int N dy = \int (2xye^{xy^2} - 3y^2) dy = e^{xy^2} - y^3, \quad x = \text{constant}.$$

The required solution is, $e^{xy^2} + x^4 - y^3 = c$, rejecting the repeated term.

Examples II(A)

Solve the following differential equations:

1. $(y + x) dx + x dy = 0.$

2. $(y + x^2) dx + x dy = 0.$

3. $y dx + (x + \cos y) dy = 0.$

4. $y dx + (x + e^y) dy = 0.$

5. $(2ax + by + g) dx + (2fy + bx + e) dy = 0.$

6. $(4x^3y^3 - 2xy) dx + (3x^4y^2 - x^2) dy = 0.$

7. $(a^2 - 2xy - y^2) dx - (x + y)^2 dy = 0.$

8. $(y^2 e^{xy^2} + 4x^3) dx + (2xye^{xy^2} - 3y^2) dy = 0.$

9. $(x^2 - 4xy - 2y^2) dx + (y^2 - 4xy - 2x^2) dy = 0.$

[C.H. 1998]

10. $(1 + 6y^2 - 3x^2y) \frac{dy}{dx} = 3xy^2 - x^2.$

11. $y(x^2 + y^2 + a^2) \frac{dy}{dx} + x(x^2 + y^2 - a^2) = 0.$

Answers

1. $x^2 + 2xy = c.$

2. $x^3 + 3xy = c$.
3. $xy + \sin y = c$.
4. $xy + e^y = c$.
5. $ax^2 + bxy + fy^2 + gx + ey = k$ (arbitrary constant).
6. $x^4y^3 - x^2y = c$.
7. $3a^2x - y^3 - 3xy^2 - x^2y = c$.
8. $e^{xy^2} + x^4 - y^3 = c$.
9. $x^3 - 6x^2y - 6xy^2 + y^3 = c$.
10. $y + 2y^3 + \frac{1}{3}x^3 - \frac{3}{2}x^2y^2 = c$.
11. $y^4 + 2a^2y^2 + x^4 - 2a^2x^2 + 2x^2y^2 = c$.

2.3 Integrating Factors

The equation

$$\tan y \, dx + \tan x \, dy = 0$$

is not exact as can be easily verified. But, if we multiply by $\cos x \cos y$, the equation becomes

$$\sin y \cos x \, dx + \sin x \cos y \, dy = 0$$

$$\text{or, } d(\sin x \sin y) = 0.$$

Consequently the primitive is $\sin x \sin y = c$.

Definition 2.3.1 A function $\mu(x, y)$ is said to be an Integrating Factor (I.F.) of the equation $M \, dx + N \, dy = 0$ if it is possible to obtain a function $u(x, y)$ such that $\mu(M \, dx + N \, dy) = du$. In other words, an I.F. is a multiplying factor by which the equation can be made exact.

The question arises as to whether or not integrating factors exist. Again, if an I.F. does exist, is it unique? These questions are answered by the following theorem:

Theorem 2.3.1 If the equation $M \, dx + N \, dy = 0$ has one and only one solution, then there exists an infinity of integrating factors.

[C.H. 1998]

Proof. Let the general solution be $f(x, y) = c$, where c is the arbitrary constant. Taking differential we obtain

$$df = \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0.$$

Since $f(x, y) = c$ is the general solution of

$$M dx + N dy = 0,$$

the relation

$$\frac{\partial f}{\partial x} / M = \frac{\partial f}{\partial y} / N$$

must hold identically.

Hence there exists a function μ such that

$$\frac{\partial f}{\partial x} = \mu M \quad \text{and} \quad \frac{\partial f}{\partial y} = \mu N.$$

Consequently, $\mu(M dx + N dy) = df$, i.e., an integrating factor μ exists.

Let $\phi(f)$ be a function of f . Then the expression

$$\mu\phi(f)(M dx + N dy) = \phi(f)df.$$

The right side is exact. So $\mu\phi(f)$ is also an I.F. Since $\phi(f)$ is an arbitrary function of f we conclude that *there exists an infinity of integrating factors*.

There are several rules for determining an I.F. in particular classes of equations. We shall discuss them in Art. 2.7. But in the present article we shall see that sometimes an I.F. can be found by inspection. A suitable regrouping of terms in the equation may be needed.

► **Example 2.3.1** Solve: $x dx + y dy + \frac{x dy - y dx}{x^2 + y^2} = 0$. [C.H. 1991]

Solution: We see that

$$x dx + y dy = \frac{1}{2} d(x^2 + y^2) \quad \text{and} \quad \frac{x dy - y dx}{x^2 + y^2} = d\left(\frac{y}{x}\right).$$

Therefore,

$$\frac{x dy - y dx}{x^2 + y^2} = \frac{\frac{x dy - y dx}{x^2}}{\frac{x^2 + y^2}{x^2}} = \frac{d(y/x)}{1 + (y/x)^2} = d\{\tan^{-1}(y/x)\}.$$

Hence the given equation becomes

$$d \left\{ \frac{1}{2} (x^2 + y^2) \right\} + d \left\{ \tan^{-1} \frac{y}{x} \right\} = 0.$$

∴ The primitive is

$$\frac{1}{2} (x^2 + y^2) + \tan^{-1}(y/x) = k \text{ (arbitrary constant)}$$

$$\text{or, } x^2 + y^2 + 2 \tan^{-1}(y/x) = c, \text{ where } c = 2k.$$

► **Example 2.3.2** Solve:

$$x \cos \left(\frac{y}{x} \right) (y \, dx + x \, dy) = y \sin \left(\frac{y}{x} \right) (x \, dy - y \, dx).$$

Solution: We note that

$$y \, dx + x \, dy = d(xy) \quad \text{and} \quad \frac{x \, dy - y \, dx}{x^2} = d \left(\frac{y}{x} \right).$$

The equation can be written as

$$\begin{aligned} \frac{x \cos \left(\frac{y}{x} \right) d(xy)}{x^2} &= y \sin \left(\frac{y}{x} \right) \frac{x \, dy - y \, dx}{x^2} \\ \text{or, } \frac{d(xy)}{xy} &= \frac{\sin \left(\frac{y}{x} \right)}{\cos \left(\frac{y}{x} \right)} d \left(\frac{y}{x} \right). \end{aligned}$$

$$\text{Integrating, } \log xy = -\log \cos(y/x) + \log c.$$

$$\therefore \log xy = \log \frac{c}{\cos(y/x)}, \quad \text{or, } xy \cos(y/x) = c.$$

Note 2.3.1 Sometimes we shall write the arbitrary constant in a suitable form so that the final expression of the primitive may not contain terms of logarithms or inverse trigonometric functions.

► **Example 2.3.3** Solve: $x \, dx + y \, dy = m(x \, dy - y \, dx).$

Solution: Dividing by $(x^2 + y^2)$, we have

$$\frac{x \, dx + y \, dy}{x^2 + y^2} = m \frac{x \, dy - y \, dx}{x^2 + y^2}, \quad \text{or, } \frac{\frac{1}{2} d(x^2 + y^2)}{x^2 + y^2} = m d \left\{ \tan^{-1} \left(\frac{y}{x} \right) \right\}.$$

$$\text{Integrating, } \frac{1}{2} \log(x^2 + y^2) = m \tan^{-1} \left(\frac{y}{x} \right) + c.$$

► **Example 2.3.4** Show that the equation

$$(x^3 - 3x^2y + 2xy^2)dx - (x^3 - 2x^2y + y^3)dy = 0$$

is exact and find the solution if $y = 1$ when $x = 1$.

[C.H. 2002]

Solution: Here $M = x^3 - 3x^2y + 2xy^2$, $N = -x^3 + 2x^2y - y^3$.

$$\therefore \frac{\partial M}{\partial y} = -3x^2 + 4xy = \frac{\partial N}{\partial x}.$$

$$\int M dx = \int (x^3 - 3x^2y + 2xy^2)dx = \frac{1}{4}x^4 - x^3y + x^2y^2,$$

$$\int N dy = \int (-x^3 + 2x^2y - y^3)dy = -x^3y + x^2y^2 - \frac{1}{4}y^4,$$

$y = \text{constant.}$

$x = \text{constant.}$

Solution: $\frac{1}{4}x^4 - x^3y + x^2y^2 - \frac{1}{4}y^4 = c$, repeated terms rejected.

When $x = 1$, $y = 1$, $c = 0$, then sol: $x^4 - 4x^3y + 4x^2y^2 - y^4 = 0$.

► **Example 2.3.5** Solve: $(xy^2 - e^{1/x^3})dx - x^2y dy = 0$. [C.H. 1998]

Solution: $xy(y dx - x dy) = e^{1/x^3}dx$. Dividing by x^4 ,

$$\frac{xy}{x^2} \cdot \frac{y dx - x dy}{x^2} = \frac{e^{1/x^3}}{x^4} dx, \text{ or, } - \int \frac{y}{x} d\left(\frac{y}{x}\right) = \int \frac{e^{1/x^3}}{x^4} dx.$$

Putting $\frac{1}{x^3} = u$,

$$-\frac{1}{2} \frac{y^2}{x^2} = -\frac{1}{3} e^{1/x^3} + c, \text{ or, } 3y^2 - 2x^2 e^{1/x^3} + 6cx^2 = 0.$$

► **Example 2.3.6** Solve: $x \frac{dy}{dx} - y = x\sqrt{x^2 + y^2}$.

[C.H. 2003]

Solution:

$$\frac{xdy - ydx}{x^2} = \sqrt{1 + \left(\frac{y}{x}\right)^2} dx, \text{ or, } \frac{d\left(\frac{y}{x}\right)}{\sqrt{1 + \left(\frac{y}{x}\right)^2}} = dx.$$

Integrating

$$\sinh^{-1}\left(\frac{y}{x}\right) = x + c, \text{ or, } y = x \sinh(x + c).$$

Below are given a table which will be helpful to the students for determining an I.F. by inspection.

Table I

	Given Expression	Integrating Factor	Exact Differential
1.	$dx \pm dy$	—	$d(x \pm y)$
2.	$x dx \pm y dy$	—	$\frac{1}{2} d(x^2 \pm y^2)$
3.	$x dy \mp y dx$	—	$d(xy)$
4.	$x dy - y dx$	$\frac{1}{x^2}$	$\frac{x dy - y dx}{x^2} = d\left(\frac{y}{x}\right)$
5.	$x dy - y dx$	$\frac{1}{x^2 + y^2}$	$\frac{x dy - y dx}{x^2 + y^2} = d\left\{\tan^{-1} \frac{y}{x}\right\}$
6.	$x dy - y dx$	$\frac{1}{xy}$	$\frac{x dy - y dx}{xy} = \frac{dy}{y} - \frac{dx}{x} = d\left\{\log \frac{y}{x}\right\}$
7.	$x dx + y dy$	$\frac{1}{x^2 + y^2}$	$\frac{x dx + y dy}{x^2 + y^2} = \frac{\frac{1}{2} d(x^2 + y^2)}{x^2 + y^2}$ $= \frac{1}{2} d\left\{\log(x^2 + y^2)\right\}$
8.	$y dx - x dy$	$\frac{1}{y^2}$	$\frac{y dx - x dy}{y^2} = d\left(\frac{x}{y}\right)$

Examples II(B)

Solve the following equations:

[Rearrange the terms, where necessary. I.F., if needed, should be found by inspection]

- $(y + x) dx + x dy = 0.$
- $y dx - x dy + \log x dx = 0.$
- $(1 + xy)y dx + (1 - xy)x dy = 0.$
- $(x + y) dy + (y - x) dx = 0.$
- $(x + y)(dx - dy) = dx + dy.$
- $x dy - y dx = \cos \frac{1}{x} dx.$
- $x dy = y(y \log x - 1) dx.$
- $x dy - y dx + a(x^2 + y^2) dx = 0.$
- $x dx + y dy + (x^2 + y^2) dy = 0.$
- $\left\{x + y \cos \left[\frac{y}{x}\right]\right\} dx = x \cos \left[\frac{y}{x}\right] dy.$

$$11. x dx + y dy = k(x dy - y dx).$$

$$12. x \frac{dy}{dx} - y = \frac{x}{a} \sqrt{(x^2 + y^2)}.$$

$$13. y(2xy + e^x) dx - e^x dy = 0.$$

$$14. (x^4 e^x - 2mxy^2) dx + 2mx^2y dy = 0.$$

$$15. (x^2 + y^2 - a^2)x dx + (x^2 - y^2 - b^2)y dy = 0.$$

$$16. (2x^2y + 4x^3 - 12xy^2 + 3y^2 - xe^y + e^{2x}) dy + (12x^2y + 2xy^2 + 4x^3 - 4y^3 + 2ye^{2x} - e^y) dx = 0.$$

$$17. x \frac{dy}{dx} - y = x\sqrt{x^2 + y^2}.$$

$$18. (xy^2 - e^{1/x^3}) dx - x^2y dy = 0.$$

$$19. \text{Show that } \frac{1}{x^2} \text{ is an integrating factor of } x dy - y dx = 0.$$

$$20. \text{Show that the equation}$$

$$(x^3 - 3x^2y + 2xy^2) dx - (x^3 - 2x^2y + y^3) dy = 0$$

is exact and find the solution if $y = 1$ when $x = 1$.

$$21. \text{Solve } (4x^3 - y^3 + 2xy^2) dx + (2x^2y - 3xy^2 + 4y^3) dy = 0; \text{ given } y = 2 \text{ when } x = 1.$$

[C.H. 2003]

[C.H. 1998]

[C.H. 2002]

Answers

$$1. x^2 + 2xy = c.$$

$$2. cx + y + \log x + 1 = 0.$$

$$3. x = cye^{1/xy}.$$

$$4. x^2 - 2xy - y^2 = c.$$

$$5. x - y = \log(x + y) + c.$$

$$6. \frac{y}{x} + \sin \frac{1}{x} = c.$$

$$7. 1 + cxy = y(1 + \log x).$$

$$8. ax + \tan^{-1} \frac{y}{x} = c.$$

9. $\frac{1}{2} \log(x^2 + y^2) + y = c.$
10. $\log x + c = \sin(y/x).$
11. $\log(x^2 + y^2) = 2k \tan^{-1}(y/x) + c.$
12. $y = x \sinh \frac{x+c}{a}.$
13. $x^2 + e^x/y = c.$
14. $x^2 e^x + m y^2 = c x^2.$
15. $x^4 - y^4 + 2x^2 y^2 - 2a^2 x^2 - 2b^2 y^2 = c.$
16. $x^2 y^2 + 4x^3 y - 4x y^3 + y^3 - x e^y + e^{2x} y + x^4 = c.$
17. $y = x \sinh(x+c).$
18. $3y^2 - 2x^2 e^{1/x^3} + 6cx^2 = 0.$
20. $x^4 - 4x^3 y + 4x^2 y^2 - y^4 = 0.$
21. $x^4 - x y^3 + x^2 y^2 + y^4 - 13 = 0.$

2.4 Equations Solvable by Separation of Variables

1. A special case of an exact equation occurs when, in the equation $M dx + N dy = 0$, M is a function of x alone [= $f_1(x)$, say] and N is a function of y alone [= $f_2(y)$, say]. The equation $f_1(x) dx + f_2(y) dy = 0$ is then said to have *separated variables*. Its general solution is

$$\int f_1(x) dx + \int f_2(y) dy = c.$$

► **Example 2.4.1** Solve: $\frac{dy}{\sqrt{1-y^2}} + \frac{dx}{\sqrt{1-x^2}} = 0.$

Solution: The general solution is clearly,

$$\begin{aligned} \sin^{-1} y + \sin^{-1} x &= c \\ \text{or, } \sin^{-1} \left\{ x\sqrt{1-y^2} + y\sqrt{1-x^2} \right\} &= c \\ \text{or, } x\sqrt{1-y^2} + y\sqrt{1-x^2} &= k, \end{aligned}$$

where $k = \sin c$ is an arbitrary constant.